Design and Implementation of a Line Following and Obstacle Avoidance Robot

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***Abstract*—....**

***Index Terms*—Line follower robot, IR sensor, Ultrasonic sensor, Arduino, Obstacle avoidance, Servo motor**

I. Introduction and Objective

A. Introduction

Automated Guided Vehicles (AGVs) are smart, driverless vehicles that bring the speed and precision of modern automation to material handling. By using advanced embedded systems to navigate safely and adapt in real time, they turn manual logistics into a predictable, data-driven backbone for Industry 4.0 [1, 2].

Line following and collision avoidance serve as the fundamental navigation and safety pillars [3]. Together, these embedded subsystems ensure the AGV can optimize space by working seamlessly alongside human workers while strictly complying with industrial safety standards like ISO 3691-4 [4, 5].

This project focuses on the systems engineering, prototyping, and development of a compact, microcontroller-driven autonomous vehicle designed for precise trajectory tracking and proactive collision mitigation [6].

B. Objective

The primary goal of this project is to implement, test, and validate a working prototype of an autonomous mobile vehicle using model-based prototyping principles. To meet the course milestones, the system must achieve the following specific technical objectives:

- **Line Following:** Use a set of infrared (IR) sensors and standard if-else logic to control the motors so the car cleanly follows a track line.
- **Obstacle Detection and Avoidance:** Integrate a pair of ultrasonic sensors to monitor front proximity metrics and implement a maneuver sequence to safely bypass detected obstructions before reacquiring the primary track line.
- **Black Zone Detection and Turnaround:** Program the car to use its floor IR sensors to spot when it hits a solid black patch on the track, stop moving forward, spin 180 degrees on the spot, and follow the line back the other way.
- **Power and Motor Integration:** Wire the hardware so that a motor controller takes a 12V battery source, drops it down to 5V to safely power the Arduino, and handles the high-current demands of the DC motors.

II. System Design and Engineering Approach

A. System Overview

The Line Following Robot is an autonomous system designed to follow a predefined path while detecting obstacles during operation. The system combines sensing, control, and actuation components to achieve reliable navigation. A systems engineering approach was applied throughout the project to identify requirements, define subsystem interactions, and develop an organized system architecture before implementation.

The robot consists of four main subsystems: the sensing subsystem, control subsystem, actuation subsystem, and power subsystem. These subsystems work together to support autonomous movement and obstacle detection.

B. System Architecture

A Block Definition Diagram (BDD) was created to represent the main components of the robot and their relationships. The diagram provides an overview of how the sensing, control, movement, and power elements are organized within the system.

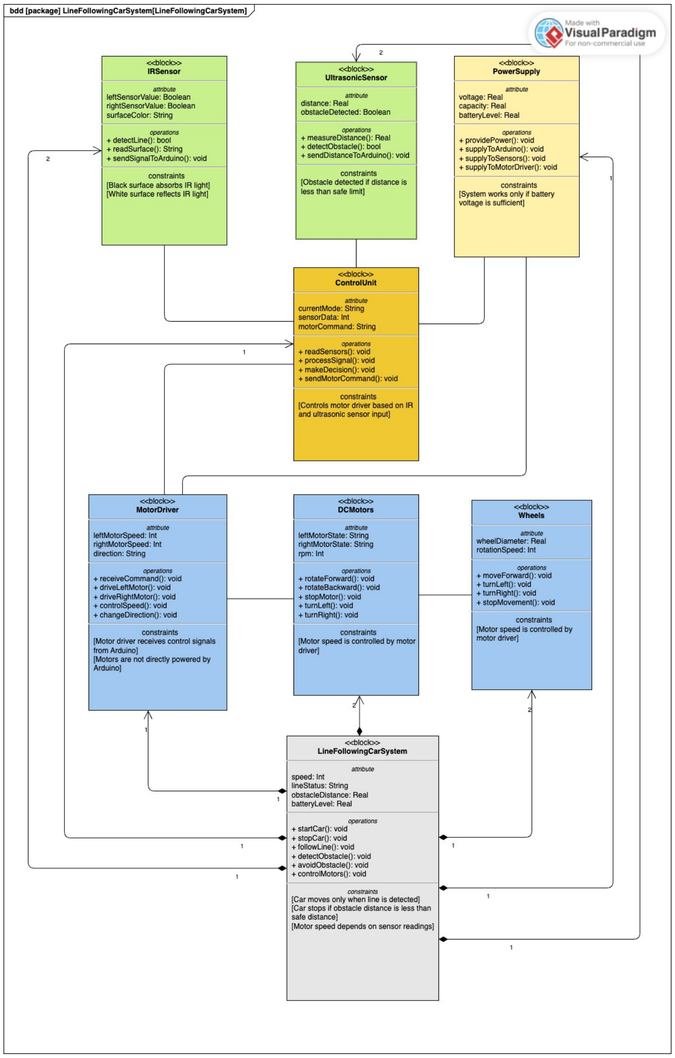

Fig. 1. Block Definition Diagram of the Line Following Robot

The sensing subsystem includes infrared (IR) sensors for line detection and ultrasonic sensors for obstacle detection. The control subsystem is based on the Arduino Uno micro-controller, which processes sensor information and determines the required motor actions. The actuation subsystem consists of the L293D motor driver and two DC motors that provide movement. The power subsystem supplies electrical energy to all hardware components.

C. Internal Component Interaction

An Internal Block Diagram (IBD) was developed to illustrate the connections and interactions between system components. The diagram shows how information and control signals are exchanged within the robot.

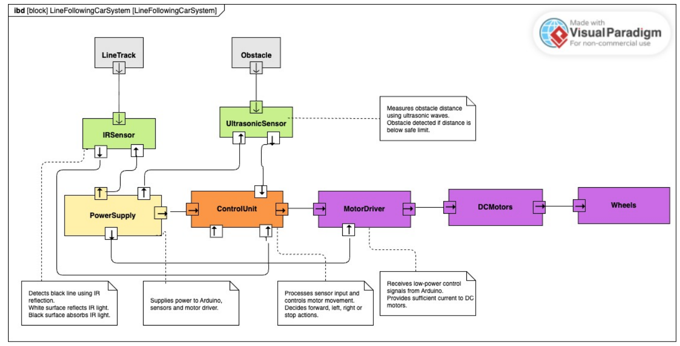

Fig. 2. Internal Block Diagram

The IR sensors continuously monitor the line position and send data to the Arduino controller. At the same time, the ultrasonic sensors measure the distance to nearby obstacles. Based on these inputs, the Arduino generates appropriate control signals for the motor driver, which adjusts the speed and direction of the motors.

D. Functional Behaviour

The behaviour of the robot was analysed using system modelling techniques. An Activity Diagram was created to describe the overall operational workflow.

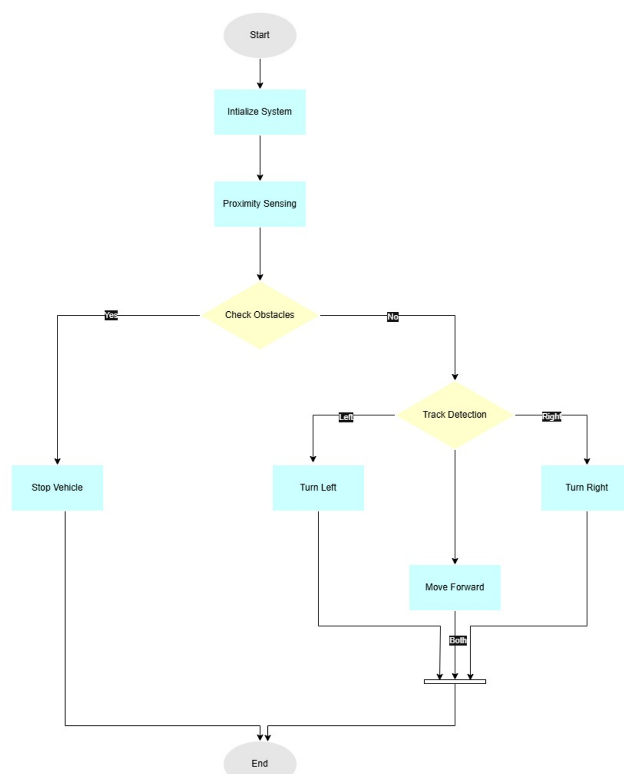

Fig. 3. Activity Diagram

The process begins with system initialization. The robot then continuously reads sensor data, evaluates the position of the line and the presence of obstacles, determines the required action, and updates motor commands. This cycle is repeated throughout operation.

A Sequence Diagram was also developed to represent the interaction between system components during robot operation.

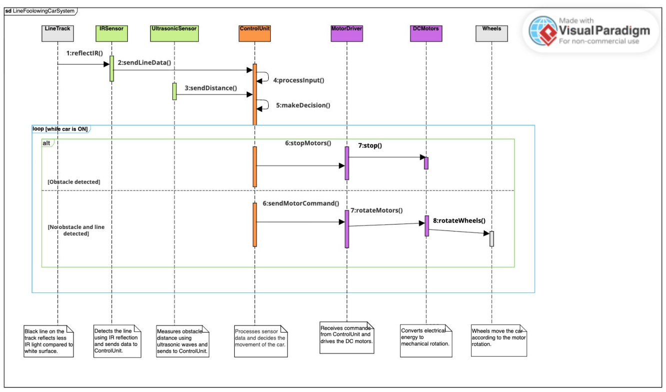

Fig. 4. Sequence Diagram

The sequence diagram illustrates how information is exchanged between the sensors, controller, and actuators during line-following and obstacle-detection activities.

E. Use Case Analysis

A Use Case Diagram was developed to identify the main functions of the robot from the user's perspective.

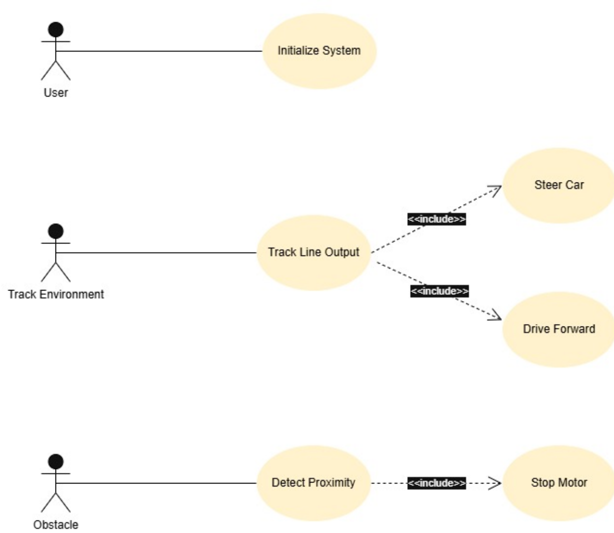

Fig. 5. Use Case Diagram

The main use cases include starting the robot, following the line, detecting obstacles, controlling movement, and stopping the robot. These functions represent the primary operational requirements of the system.

F. System States

A State Machine Diagram was created to model the different operational states of the robot.

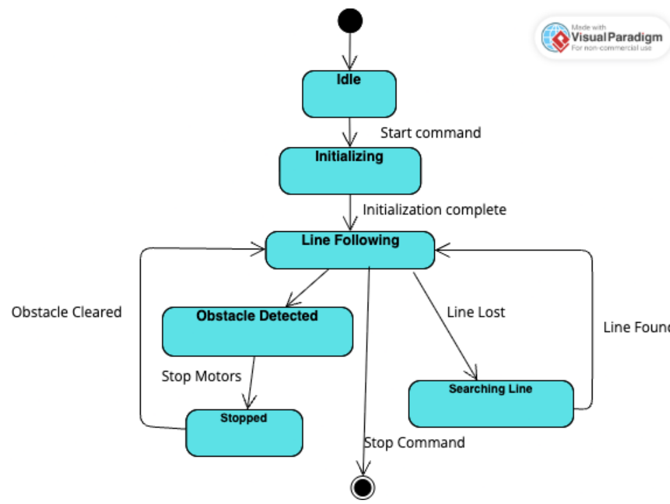

Fig. 6. State Machine Diagram

The robot initially remains in the Idle state. Once activated, it transitions to the Line Following state. When an obstacle is detected, the system moves to the Obstacle Detected state and performs a stop or avoidance action. After the obstacle is cleared, the robot returns to normal operation.

G. Design Decisions

Several design decisions were made during the development of the prototype. The Arduino Uno was selected because it is easy to program, widely supported, and compatible with various sensors and motor drivers. IR sensors were chosen for accurate line detection, while ultrasonic sensors were used to provide reliable obstacle detection. The L293D motor driver was selected because it allows simple and effective control of two DC motors.

The use of SysML diagrams improved system understanding by providing both structural and behavioural views of the robot. These models supported the implementation process and helped ensure consistency between system requirements and the final prototype design.

III. Hardware Software Implementation

IV. Simulation Testing Results

V. Challenges and Solutions

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VI. Conclusion and Future Improvements

A. Conclusion

This project successfully demonstrated the design and development of a Line Following Robot capable of autonomous navigation and obstacle detection. By integrating infrared sensors, ultrasonic sensors, an Arduino Uno microcontroller, and a motor control system, the robot was able to follow a predefined path while responding to environmental conditions.

A systems engineering approach was applied throughout the project to support requirement identification, system modelling, implementation, and testing. The use of SysML diagrams provided a clear representation of both the structural and behavioural aspects of the system, helping to improve understanding of component interactions and overall system functionality.

The testing results showed that the robot was able to perform the intended line-following and obstacle-detection tasks under normal operating conditions. Therefore, the project achieved its primary objective of developing a functional robotic prototype while applying key systems engineering principles.

B. Future Improvements

Although the prototype met the project requirements, several improvements could be implemented in future developments. One possible enhancement is the integration of a PID control algorithm to improve movement accuracy and provide smoother line-following performance. Additional infrared sensors could also be incorporated to increase detection reliability, particularly when navigating sharp curves or complex paths.

The mechanical design could be further optimized by improving chassis stability, wheel alignment, and component placement. These modifications may contribute to better overall performance and durability of the robot.

Furthermore, wireless communication technologies such as Bluetooth or Wi-Fi could be added to enable remote monitoring and control. Future versions may also incorporate camera-based vision systems or machine learning techniques to support more advanced navigation and obstacle-avoidance capabilities.

Overall, the proposed improvements would increase the robot's accuracy, adaptability, and functionality, making it more suitable for complex autonomous applications.